Appendix A

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1. Show that the definition of rational equivalence in §1 is equivalent to the equivalence relation generated by the following relation: two cycles Y, Z of codimension r on X are equivalent if there exists a cycle W of codimension r on $X \times \mathbb{A}^1$ which intersects $X \times \{0\}$ and $X \times \{1\}$ properly, and such that $Y = W.(X \times \{0\})$, $Z = W.(X \times \{1\})$.

2. Prove the following result about Weil divisors, which generalizes (IV, Ex. 2.6), and which is needed to show that f_* is well-defined modulo ratonal equivalence (A3): Let $f : X \to X'$ be a proper, generically finite map of normal varieties, and let D_1 and D_2 be linearly equivalent Weil divisors on X. Then f_*D_1 and f_*D_2 are linearly equivalent Weil divisors on X'.

3. Show directly that any subvariety of degree d in $\mathbb{P}^n$ is rationally equivalent to d times a linear space of the same dimension, by using a projection argument similar to (III, 9.8.3).

4. Let $\pi : X \to C$ be a ruled surface (V, §2) over a nonsingular curve C. Show that the group $A^2(X)$ of zero-cycles modulo rational equivalence is isomorphic to $\operatorname{Pic} C$.

5. Let X be a surface, let $P \in X$ be a point, and let $\pi : \tilde{X} \to X$ be the monoidal transformation with center P (V, §3). Show that $A(\tilde{X}) \cong \pi^* A(X) \oplus \mathbb{Z}$, where $\mathbb{Z}$ is generated by the exceptional curve $E \in A^1(X)$, and the intersection theory is determined by $E^2 = -\pi^*(P)$.

6. Let X be a nonsingular projective variety of dimension n, and let $\Delta \subset X \times X$ be the diagonal. Show that $c_n(\mathcal{T}_X) = \Delta^2$ in $A^n(X)$, under the natural isomorphism of X with Δ.

Proof. □

7. Let X be a nonsingular projective 3-fold, with Chern classes c_1, c_2, c_3. Show that

$$1 - p_a = \frac{1}{24} c_1 c_2,$$

and for any divisor D,

$$\chi(\mathcal{O}_X(D)) = \frac{1}{12} D.(D - K).(2D - K) + \frac{1}{12} D.c_2 + 1 - p_a.$$

Proof. Let $\mathcal{E} = \mathcal{O}_X(D)$ for some divisor D. Then

$$\operatorname{ch}(\mathcal{E}) = 1 + D + \frac{1}{2} D^2 + \frac{1}{6} D^3.$$

We denote by c_1, c_2, and c_3 the Chern classes of the tangent sheaf $\mathcal{T}_X$. Since $\mathcal{T}_X$ is the dual of Ω_X, and since $c_1(\Omega_X) = c_1(\wedge^3 \Omega_X)$ by (C5), and since $\wedge^3 \Omega_X = \omega_X$ is just $\mathcal{O}_X(K)$, where K is the canonical divisor, we see that $c_1(\mathcal{T}_X) = -K$. Using $c_1 = -K$ and c_2, c_3, we have

$$\operatorname{td}(\mathcal{T}_X) = 1 - \frac{1}{2} K + \frac{1}{12}(K^2 + c_2) - \frac{1}{24} K.c_2.$$

We can write (4.1) as

$$\chi(\mathcal{O}_X(D)) = \deg \left(\left(1 + D + \frac{1}{2} D^2 + \frac{1}{6} D^3 \right) . \left(1 - \frac{1}{2} K + \frac{1}{12}(K^2 + c_2) - \frac{1}{24} K.c_2 \right) \right)_3$$

$$= \frac{1}{6}D^3 - \frac{1}{4}D^2.K + \frac{1}{12}D.(K^2 + c_2) - \frac{1}{24}K.c_2$$
$$= \frac{1}{12}(2D^3 - 3D^2.K + D.K^2) + \frac{1}{12}D.c_2 - \frac{1}{24}K.c_2$$
$$= \frac{1}{12}D.(D-K).(2D-K) + \frac{1}{12}D.c_2 - \frac{1}{24}K.c_2.$$

In particular, for $D = 0$, we find that

$$\chi(\mathcal{O}_X) = -\frac{1}{24}K.c_2 = \frac{1}{24}c_1.c_2,$$

and by definition of the arithmetic genus (III, Ex. 5.3), this says

$$1 - p_a = \frac{1}{24}c_1.c_2.$$

□

8. Let $\mathcal{E}$ be a locally free sheaf of rank 2 on $\mathbb{P}^3$, with Chern classes c_1, c_2. Since $A(\mathbb{P}^3) \cong \mathbb{Z}[h]/(h^4)$, we can think of c_1 and c_2 as integers. Show that $c_1c_2 \equiv 0 \mod 2$.

Proof. The exponential Chern character of $\mathcal{E}$ is given by

$$\mathrm{ch}(\mathcal{E}) = 2 + c_1 + \frac{1}{2}(c_1^2 - 2c_2) + \frac{1}{6}(c_1^3 - 3c_1c_2)$$

Following the computation in (4.1.3), the Chern polynomial of $\mathcal{T}_{\mathbb{P}^3}$ is given by

$$c_t(\mathcal{T}_{\mathbb{P}^3}) = (1+ht)^4 = 1 + 4ht + 6h^2t^2 + 4h^3t^3,$$

so the Todd class of $\mathcal{T}_{\mathbb{P}^3}$ is given by

$$\mathrm{td}(\mathcal{T}_{\mathbb{P}^3}) = 1 + 2h + \frac{11}{6}h^2 + h^3,$$

where we substituted $K = -4h$. By (4.1), we have

$$\chi(\mathcal{E}) = \deg\left(\left(2 + c_1 + \frac{1}{2}(c_1^2 - 2c_2) + \frac{1}{6}(c_1^3 - 3c_1c_2)\right).\left(1 + 2h + \frac{11}{6}h^2 + h^3\right)\right)_3$$
$$= 2 + \frac{11}{6}c_1 + c_1^2 - 2c_2 + \frac{1}{6}c_1^3 - \frac{1}{2}c_1c_2$$
$$= 1 + \binom{c_1+3}{3} - 2c_2 - \frac{1}{2}c_1c_2.$$

A priori $\chi(\mathcal{E})$ and every term above besides $\frac{1}{2}c_1c_2$ is an integer, so $\frac{1}{2}c_1c_2$ must be an integer as well. Hence, $c_1c_2 \equiv 0 \mod 2$. □

9. *Surfaces in* $\mathbb{P}^4$

(a) Verify the formula of (4.1.3) for the rational cubic scroll in $\mathbb{P}^4$ (V, 2.19.1).

(b) If X is a K3 surface in $\mathbb{P}^4$, show that its degree must be 4 or 6.

(c) If X is an Abelian surface in $\mathbb{P}^4$, show that its degree must be 10.

(d) Determine which of the rational ruled surfaces X_e, $e \geq 0$ (V, §2), admit an embedding in $\mathbb{P}^4$.

Proof.

(a) The rational cubic scroll $X \subset \mathbb{P}^4$ is a surface of degree $d = 3$ and invariant $e = 1$, and it is embedded in $\mathbb{P}^4$ via the complete linear system associated to the divisor $H = C_0 + 2f$, where C_0 is the divisor class of a section such that $C_0^2 = -e = -1$ and f is the divisor class of a fiber. The canonical divisor is linearly equivalent to $K \sim -2C_0 - 3f$ (2.11), so that

$$K^2 = 8, \quad H.K = -5,$$

and it has arithmetic genus $p_a(X) = 0$ (V, 2.5). Combining everything,

$$d^2 - 10d - 5H.K - 2K^2 + 12 + 12p_a = 9 - 30 + 25 - 16 + 12 + 0 = 0,$$

so the formula of (4.1.3) indeed holds.
We compute some other invariants. The second Chern class of X can be computed using (4.1),

$$1 + p_a(X) = \frac{1}{12}(K^2 + c_2) \quad \Longrightarrow \quad c_2 = 4.$$

Let $\mathcal{N}$ be the normal sheaf of X in $\mathbb{P}^4$. Then, we have

$$c_1(\mathcal{N}) = 5H + K = 3C_0 + 7f$$
$$c_2(\mathcal{N}) = 10H^2 - c_2 + 5H.K + K^2 = 29.$$

(b) If X is a K3 surface, then $K \sim 0$ and $p_a(X) = 1$. Plugging these values into the formula of (4.1.3) gives the quadratic polynomial

$$d^2 - 10d + 24 = 0,$$

so we must have $d = 4$ or 6.

(c) The arithmetic genus of a K3 surface is $p_a(X) = -1$, and since H is very ample and $12K \sim 0$, we have $H.K = 0$ and $K^2 = 0$. The formula of (4.1.3) gives

$$d^2 - 10d = 0 \quad \Longrightarrow \quad d = 10.$$

□

10. Use the fact that the tangent sheaf of an abelian variety is free, to show that it is impossible to embed an abelian 3-fold in $\mathbb{P}^5$.

Proof. Let X be an abelian 3-fold, and suppose there exists an embedding $i : X \hookrightarrow \mathbb{P}^5$ of degree d. In the Chow ring of $\mathbb{P}^5$, X is equivalent to d times a 3-plane h^2, where $h \in A^1(\mathbb{P}^5)$ is the class of a hyperplane, so $X.X = d^2h^4$. On the other hand, the self-intersection formula (C5) says

$$i_*(c_2(\mathcal{N})) = X.X,$$

where $\mathcal{N}$ is the normal sheaf of X in $\mathbb{P}^5$. There is an exact sequence

$$0 \longrightarrow \mathcal{T}_X \longrightarrow i^*\mathcal{T}_{\mathbb{P}^5} \longrightarrow \mathcal{N} \longrightarrow 0 \ .$$

We use this sequence and the fact that $\mathcal{T}_X \cong \mathcal{O}_X^{\oplus 3}$ to compute $c_2(\mathcal{N})$, and thus get a contradiction.
First, we use the exact sequence

$$0 \longrightarrow \mathcal{O}_{\mathbb{P}^5} \longrightarrow \mathcal{O}_{\mathbb{P}^5}(1)^{\oplus 6} \longrightarrow \mathcal{T}_{\mathbb{P}^5} \longrightarrow 0 \ ,$$

so we see that

$$c_t(\mathcal{T}_{\mathbb{P}^5}) = (1 + ht)^6 = 1 + 6ht + 15h^2t^2 + \cdots .$$

On the other hand,

$$c_t(\mathcal{T}_X) = c_t(\mathcal{O}_X^{\oplus 3}) = c_t(\mathcal{O}_X)^3 = 1.$$

Therefore, denoting by $H \in A^1(X)$ the class of a hyperplane section of X, we have from the exact sequence above, using (C3), that

$$c_t(\mathcal{N}) = 1 + 6Ht + 15H^2t^2$$

Comparing coefficients of t^2, we find that

$$c_2(\mathcal{N}) = 15H^2 \quad \Longrightarrow \quad i_*c_2(\mathcal{N}) = 15h^4.$$

But $d^2 = 15$ is impossible. Hence, X cannot be embedded in $\mathbb{P}^5$. □